

RESEARCH ARTICLE

Comparison between supervised and physics-informed unsupervised deep neural networks for estimating cerebral perfusion using multi-delay arterial spin labeling MRI

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Funding information

Japan Society for the Promotion of Science, Grant/Award Numbers: 21K07616, 21K15802

Abstract

This study aimed to implement a physics-informed unsupervised deep neural network (DNN) to estimate cerebral blood flow (CBF) and arterial transit time (ATT) from multi-delay arterial spin labeling (ASL), and compare its performance with that of a supervised DNN and the conventional method. Supervised and unsupervised DNNs were trained using simulation data. The accuracy and noise immunity of the three methods were compared using simulations and in vivo data. The simulation study investigated the differences between the predicted and ground-truth values and their variations with the noise level. The in vivo study evaluated the predicted values from the original images and noise-induced variations in the predicted values from the synthesized noisy images by adding Rician noise to the original images. The simulation study showed that CBF estimated using the supervised DNN was not biased by noise, whereas that estimated using other methods had a positive bias. Although the ATT with all methods exhibited a similar behavior with noise increase, the ATT with the supervised DNN was less biased. The in vivo study showed that CBF and ATT with the supervised DNN were the most accurate and that the supervised and unsupervised DNNs had the highest noise immunity in CBF and ATT estimations, respectively. Physics-informed unsupervised learning can estimate CBF and ATT from multi-delay ASL signals, and its performance is superior to that of the conventional method. Although noise immunity in ATT estimation was superior with unsupervised learning, other performances were superior with supervised learning.

KEYWORDS

arterial spin labeling (ASL), arterial transit time (ATT), cerebral blood flow (CBF), supervised learning, unsupervised learning

Abbreviations: ASL, arterial spin labeling; ATT, arterial transit time; CBF, cerebral blood flow; CV_{Net}, normalized coefficient of variation over repeated training; DNN, deep neural network; GM, gray matter; KM, kinetic model; LD, labeling duration; LOA, limit of agreement; LS, least squares; MAE, mean absolute error; MSE, mean squared error; N_{HL}, number of hidden layers; N_{NR}, number of neurons; NMAE, normalized mean absolute error; NRMSE, normalized root mean squared error; PILF, physics-informed loss function; PLD, post-labeling delay; WD, signal-weighted delay; WGM, whole gray matter; WWM, whole white matter.

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1 | INTRODUCTION

Arterial spin labeling (ASL) MRI can noninvasively quantify cerebral blood flow (CBF) and arterial transit time (ATT), which is the travel time of labeled spins from the labeling plane to the brain tissue, using magnetically inverted spins in arterial blood as diffusible endogenous tracers.¹⁻³ Although the clinical utility of CBF and ATT has been widely reported,⁴⁻⁸ the intrinsically poor signal-to-noise ratio (SNR) of ASL compromises their accuracy and precision^{9,10} and may undermine the widespread use of ASL in clinical practice. Thus, accurate and precise parameter estimation techniques are required to mitigate the poor SNR of ASL.

Conventionally, multiple averaging is the solution employed for the low SNR of ASL. Other conventional approaches to improve the accuracy of parameter estimation include optimization of the acquisition design and implementation of sophisticated parameter estimation techniques.¹⁰⁻¹⁶ Recently, deep learning (DL) has been widely used in ASL.¹⁷⁻²¹ A simulation-based deep neural network (DNN) has shown higher accuracy and noise immunity than a conventional method that uses the signal-weighted-delay (WD) method and the least-squares (LS) solution for estimating ATT and CBF, respectively.²¹ This DNN adopted a supervised training method. Physics-informed unsupervised learning is another training approach that incorporates the physical model into the loss function, with documented performance reports in other fields.^{22,23} In this approach, the objective of network training is to minimize the discrepancy between predicted signals, derived from the physics model integrated into the loss function, and the actual measured signals. However, it has not been introduced to ASL. These two learning methods have different advantages. A major advantage of supervised learning is that it can learn from the ground truth²¹; in contrast, physics-informed unsupervised learning can be trained directly using measured data without any ground truth.²³

Physics-informed unsupervised learning, which incorporates a kinetic model (KM) of ASL based on the modified Bloch equation,²⁴⁻²⁶ may improve the performance of DL-based parameter estimation because it does not require any prior information, owing to the property of direct learning from measured signals, whereas supervised learning requires a priori information (for example, noise distribution) and can only learn from generally known information. In addition, direct training can reveal several ASL signal characteristics, including noise distribution, which may allow for the learning of latent features of ASL that have not been reported previously. Although physics-informed unsupervised learning does not require any prior information but requires abundant data measured under different conditions, obtaining ASL images of various diseases and wide noise levels is difficult. Therefore, this study used simulation data to train DNNs^{22,23} because simulation can generate training data without any restrictions and under arbitrary conditions such as physiological parameters (i.e., CBF and ATT) and noise levels. The objectives of this study were to (1) formulate a physics-informed loss function (PILF) incorporating a KM of ASL, (2) compare its performance with supervised learning and the conventional method, and (3) clarify the parameter-estimation characteristics of each learning method.

2 | METHODS

First, unsupervised and supervised DNNs were trained and optimized using simulation data. Subsequently, the performances of unsupervised learning, supervised learning, and the conventional method were compared using the simulation and in vivo data. All processing was performed using MATLAB 2022a (MathWorks, Natick, MA, USA) and a personal computer with an Intel Core i9-10980XE CPU at 3.00 GHz and an NVIDIA GeForce RTX 3090. In particular, all DNNs were implemented using the Deep Learning Toolbox in MATLAB.

2.1 | ASL acquisition

A combination of the Hadamard-encoding scheme with extra encoding and single-delay acquisition with a long labeling duration (LD) and long post-labeling delay (PLD)¹⁰ resulted in five delays: (LD [ms], PLD [ms]) = (1333, 700), (1333, 2033), (1333, 3367), (4000, 700), and (4000, 3000). This bolus design was employed for both the simulation and in vivo experiments. The actual scan conditions related to the readout sequence employed for the in vivo scan are outlined in Section 2.7.

2.2 | Generating simulation data

First, theoretical ASL signals were calculated using the single-compartment model (Equation 1), in which the ground truth ranges of the CBF and ATT were 0–120 mL/100 g/min and 700–3367 ms (i.e., minimum to maximum PLD), respectively. In total, 30 000, 500, and 5000 CBF and ATT pairs were uniformly and randomly generated for the training, validation, and test datasets, respectively.

$$\frac{\Delta M}{M_0} = \frac{2\alpha\beta T_{1t}}{\lambda} f \exp\left(\frac{-\delta}{T_{1a}}\right) \left[\exp\left(\frac{-\max(w-\delta, 0)}{T_{1t}}\right) - \exp\left(\frac{-\max(\tau+w-\delta, 0)}{T_{1t}}\right) \right] \quad (1)$$

where ΔM is the ASL signal, M_0 is the equilibrium magnetization of brain tissue, α is the labeling efficiency (0.85), β is the background suppression efficiency (0.75), λ is the blood partition coefficient (0.9), f is the CBF, T_{1t} is the T_1 of brain tissue (1.2 s), T_{1a} is the T_1 of arterial blood (1.66 s), δ is the ATT, w is the PLD, and τ is the LD.

Second, Rician noise was added to the theoretical ASL signals using the method reported in a previous study.²¹ The ASL signals were assumed to comprise four components corresponding to real (R) and imaginary (I) parts of the control (C) and label (L) signals and were divided into four components (i.e., C_R , C_I , L_R , and L_I). Subsequently, independent and identical Gaussian noise with an SD of 0%–0.5% M_0 was added to each component, where 100, 100, and 51 noise levels spaced evenly were used for the training, validation, and test datasets, respectively. The magnitude signals were calculated using the real and imaginary signals to simulate Rician noise ($C = \sqrt{C_R^2 + C_I^2}$ and $L = \sqrt{L_R^2 + L_I^2}$), and they were summed to simulate the subtraction process of ASL ($C + L$). Finally, the training, validation, and test datasets comprised 3 000 000, 50 000, and 255 000 data points, respectively.

2.3 | Unsupervised learning

PILFs evaluate the consistency between the observed signal (i.e., inputs of DNN) and the predicted signal, which is calculated from the predicted CBF ($\hat{f}$) and ATT ($\hat{\delta}$) using the single-compartment model. Subsequently, consistency was assessed as differences between the input and predicted signals using the mean absolute error (MAE) and mean squared error (MSE) and minimized by training. The PILFs using MAE and MSE were referred to as MAE_{KM} and MSE_{KM}, respectively (<FIG 1>Figure 1).

$$\text{Loss}_{\text{MAE}} = \frac{1}{n} \sum_{n \in N_{\text{PLD}}} \left| \left(\frac{\Delta M}{M_0} \right)_{\text{Pred}} - \left(\frac{\Delta M}{M_0} \right)_{\text{Input}} \right| \quad (2)$$

$$\text{Loss}_{\text{MSE}} = \frac{1}{n} \sum_{n \in N_{\text{PLD}}} \left[\left(\frac{\Delta M}{M_0} \right)_{\text{Pred}} - \left(\frac{\Delta M}{M_0} \right)_{\text{Input}} \right]^2 \quad (3)$$

where $(\Delta M/M_0)_{\text{Pred}}$ is the normalized ASL signal calculated from the predicted CBF ($\hat{f}$) and ATT ($\hat{\delta}$) using the single-compartment model, $(\Delta M/M_0)_{\text{Input}}$ is the input signal, and N_{PLD} is the number of PLDs.

A DNN consisting of two subnetworks with the same number of hidden layers (N_{HL}) and neurons (N_{NR}) was implemented to simultaneously predict CBF and ATT, because the predicted CBF and ATT values should be concurrently passed on to the PILF (Figure 1). The DNN has an input layer, fully connected layers with the same number of neurons, an exponential linear unit activation function,²⁷ an output layer, an Adam optimizer,²⁸ and the Kaiming He initialization.²⁹ The DNN was optimized for N_{HL} (3, 5, 7, and 9), N_{NR} (5, 50, 100, and 150), learning rate (10^{-3} , 10^{-4} , and 10^{-5}), and loss functions (MAE_{KM} and MSE_{KM}), that is, an optimized setting was selected from 96 patterns of configurations using a grid search.

2.4 | Supervised learning

Based on a previous report, two individual DNNs were implemented to estimate CBF and ATT because of the superior performance of the DNNs compared with a single DNN that simultaneously predicts CBF and ATT.²¹ Supervised DNNs comprise an input layer, fully connected layers with the same number of neurons, an exponential linear unit activation function,²⁷ an output layer, an Adam optimizer,²⁸ and the Kaiming He initialization.²⁹ Similar to the unsupervised DNNs, N_{HL} (3, 5, 7, and 9), N_{NR} (5, 50, 100, and 150), learning rate (10^{-3} , 10^{-4} , and 10^{-5}), and loss functions (MAE and MSE) were optimized. These candidate hyperparameters differed from those used in a previous study because we aligned them with those of unsupervised learning.²¹ Notably, in both learning methods, if the validation loss did not improve for 10 consecutive iterations, the training session was terminated before the scheduled setting.

2.5 | Conventional method

The conventional method¹⁰ estimates ATT using the WD method¹² and CBF using the LS solution. The WD values were calculated from the observed signals and converted to ATT by numerical comparison with the theoretical WD values.

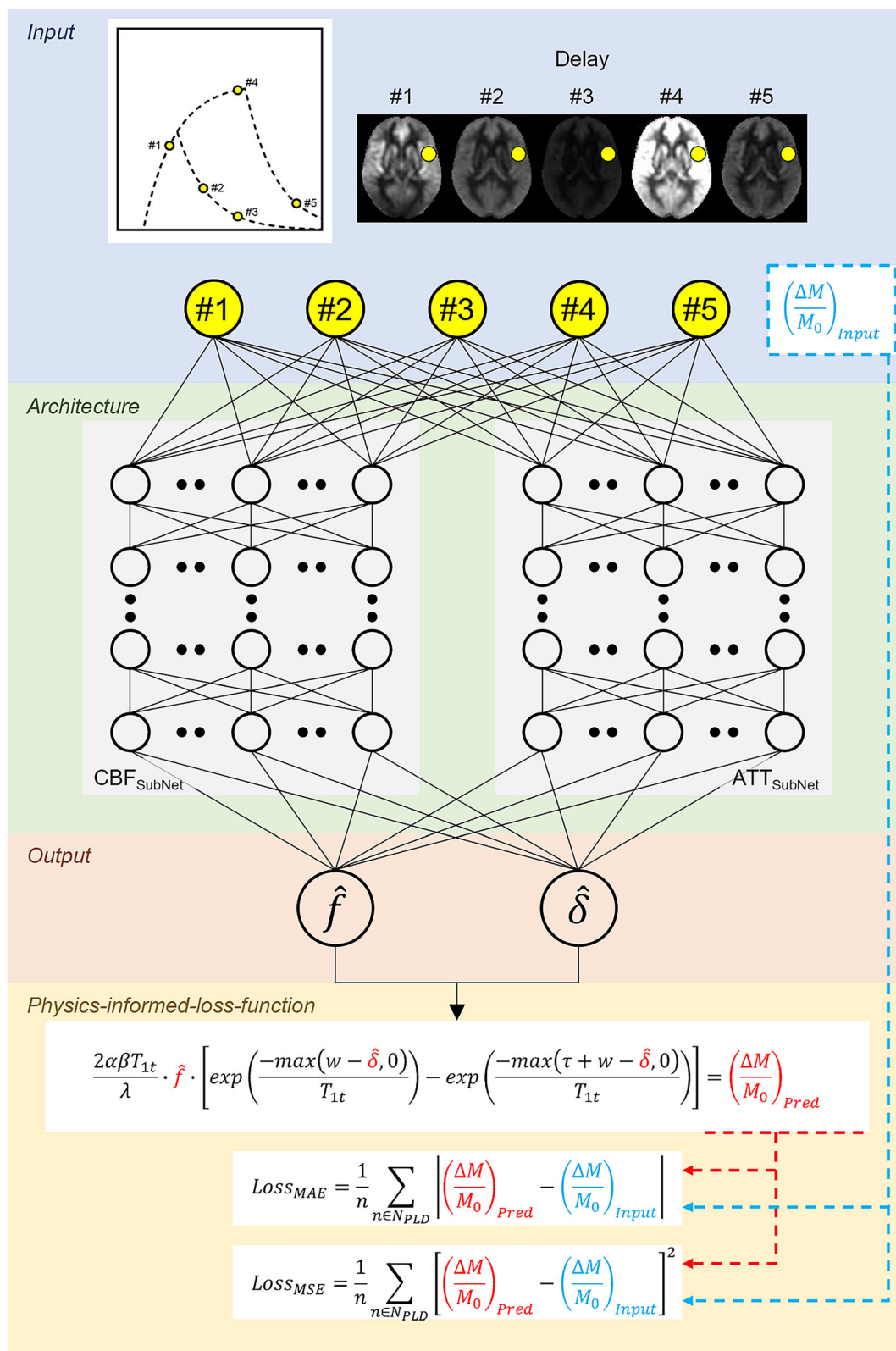


FIGURE 1 Overview of training process of the PILF and schematic diagram of the unsupervised network. First, a network consisting of two subnetworks simultaneously predicts CBF ($\hat{f}$) and ATT ($\hat{\delta}$) from the input signals, $(\Delta M/M_0)_{Input}$. Subsequently, ASL signals, $(\Delta M/M_0)_{Pred}$, are calculated using the single-compartment model with the predicted parameters. Finally, the consistency between $(\Delta M/M_0)_{Input}$ and $(\Delta M/M_0)_{Pred}$ is evaluated using MAE ($Loss_{MAE}$) or MSE ($Loss_{MSE}$) and minimized by the training process.

$$WD(\delta) = \frac{\left[\sum_{i=1}^n w_i \Delta M(\delta, w_i) \right]}{\left[\sum_{i=1}^n \Delta M(\delta, w_i) \right]} \quad (4)$$

where ΔM is the ASL signal, δ is the ATT, w_i is the i th PLD, and n is the number of delays.

Subsequently, CBF was calculated using the WD-derived ATT and the LS solution.

$$\hat{f} = (A^T A)^{-1} A^T B \quad (5)$$

where $\hat{f}$ is the LS solution (CBF) of the linear problem $B = Af$, A is a column vector of the coefficients (C) of each bolus from Equation (1) except for f ($A \triangleq [C_1 \cdots C_n]^T$), B is a column vector of ASL signals ($B \triangleq [\Delta M_1 \cdots \Delta M_n]^T$), and n is the number of delays.

2.6 | Evaluation in simulation data

Network optimization and comparisons among the methods were performed using a test dataset based on the normalized mean absolute error (NMAE), normalized root mean squared error (NRMSE), and normalized coefficient of variation over repeated training (CV_{Net}).

NMAE and NRMSE evaluate the accuracy of the predicted values by comparing them with the ground truth values.

$$NMAE = \frac{1}{\bar{x}_{GT}} \left(\frac{1}{n} \sum_{i=1}^n |x_i^{GT} - x_i^{Pred}| \right) \quad (6)$$

$$NRMSE = \frac{1}{\bar{x}_{GT}} \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i^{GT} - x_i^{Pred})^2} \quad (7)$$

where $\bar{x}_{GT}$ is the mean of the ground truth, n is the number of data points, x_i^{GT} is the i th ground truth, and x_i^{Pred} is the i th predicted value.

CV_{Net} assesses network precision by measuring the fluctuation of estimators during repeated training because training is a stochastic process,²³ whereas the conventional method is deterministic (i.e., CV_{Net} is zero).

$$CV_{Net} = \frac{1}{\bar{x}_{GT}} \sqrt{\frac{1}{n \times m - 1} \sum_{i=1}^n \sum_{j=1}^m (x_{ij} - \bar{x}_i)^2} \quad (8)$$

where $\bar{x}_{GT}$ is the mean of the ground truth, n is the number of data points, m is the number of repeated training sessions (20), x_{ij} is the j th repeated prediction of the i th data point, and $\bar{x}_i$ is the mean of the repeated predictions of the i th data point.

The DNN configurations (N_{HL} , N_{NR} , learning rate, and loss functions) were preliminarily optimized for unsupervised and supervised DNNs via ranking,^{21,23} where the respective metrics were ranked among 96 configurations. Six rankings ($NMAE_{CBF}$, $NRMSE_{CBF}$, CV_{CBF} , $NMAE_{ATT}$, $NRMSE_{ATT}$, and CV_{ATT}) were summed to determine the optimized configuration, and the lowest summed ranking was considered the optimized setting.

After the optimization of each DNN, the estimation performance was compared among unsupervised DNN, supervised DNN, and the conventional method by evaluating the changes in NMAE, NRMSE, and CV_{Net} with the noise level. Bland–Altman analyses were performed between the ground truth and predicted values at three noise levels (SD of 0%, 0.02%, and 0.20% M_0) representatively. Moreover, using the same data, the relationship between the prediction accuracy of each predicting parameter and ground truth pair of CBF and ATT was evaluated to determine whether the degraded accuracy originated from CBF or ATT being predicted, because a pair of ground truth CBF and ATT values defines a signal series during the generation of simulation signals. In other words, the estimation performance can be affected by both the CBF and ATT. The prediction accuracy was calculated as the percent difference (the difference between the predicted and ground truth values divided by the ground truth value).

2.7 | In vivo data acquisition

After approval of the institutional review board, 17 healthy adults were enrolled (five females and 12 males; age range 22–58, years; average age 33 years). Informed consent was obtained from all the participants. A 3.0 T MRI unit (Discovery MR750; GE Healthcare, Waukesha, WI, USA) with a 32-channel head coil was used to acquire multi-delay ASL with the aforementioned bolus design and T_1 -weighted images. ASL with a 3D fast spin-echo stack of spirals was performed using a repetition time (TR) of 6225 and 8525 ms for Hadamard-encoding and single-delay scans, respectively; echo time (TE) of 10.5 ms; field-of-view (FOV) of 240 mm; slice thickness of 4 mm; 512 points with six interleaves; and one average. A 3D inversion-recovery prepared spoiled gradient recalled echo was performed to acquire 3D T_1 -weighted images using a TR of 6.2 ms, TE of 2.1 ms, slice thickness of 1 mm, FOV of 240×216 mm, flip angle of 15° , and inversion time (TI) of 500 ms. It should be noted that all images were the same as those in previous studies^{10,21} and were retrospectively analyzed in this study.

2.8 | Evaluation with in vivo data

To synthesize noisy ASL images, Rician noise was added to the original ASL images at six SDs (0%–0.1% M_0 , 0.02 intervals) using the same method as that used in the simulation analysis. The CBF and ATT maps were computed from the original and noisy images using the three methods. The computed maps were then coregistered with individual 3D- T_1 images and spatially normalized to the Montreal Neurological Institute space template using SPM12 (Wellcome Center for Human Neuroimaging, UCL). CBF and ATT were measured in whole gray matter (WGM) and whole white matter (WWM) using tissue masks created from individual 3D- T_1 images with a probability threshold of 75%. CBF and ATT values were measured in the maps predicted from the original images and were statistically compared. The relative changes in CBF and ATT with the added noise levels were also evaluated to compare the noise immunity.

In addition to the simulation-based DNNs, we trained an unsupervised DNN with the same configurations as the optimized settings using the original and synthesized in vivo images ($n = 15$ out of 17; 4 females and 11 males; age range 22–58, years; average age 34 years). Then, the performances of the four methods (i.e., unsupervised DNN trained using in vivo data, unsupervised DNN trained using simulation data, supervised DNN, and conventional method) were compared using a test dataset ($n = 2$ out of 17; a 22-year-old male and a 25-year-old female). To prepare the training data, we first coregistered 3D- T_1 images with ASL images to avoid unnecessary transformations of the ASL signals. Then, tissue segmentation was performed, and gray matter (GM) masks were created using a conservative probability threshold of 95% because GM (i.e., $T_{1t} = 1.2$ s) was assumed during the quantification process. Finally, the original and synthesized noisy images were masked to extract the signal series, and an unsupervised DNN was trained (Table S1 and Figure S1).

2.9 | Statistical analysis

GraphPad Prism 9.3 (GraphPad Software, La Jolla, CA, USA) was used for statistical analysis. The in vivo CBF and ATT values predicted from the original images were compared among the methods using one-way analysis of variance with a post hoc Tukey' multiple comparisons test. Statistical significance was defined at $p < 0.05$.

3 | RESULTS

Table 1 shows the optimized hyperparameters of the unsupervised and supervised DNNs defined according to the preliminary grid search. The subsequent results were obtained using the optimized DNNs.

TABLE 1 Optimized hyperparameters of each network.

Hyperparameters	Unsupervised	Supervised CBF	Supervised ATT
N_{HL}	3	9	9
N_{NR}	150	50	100
LR	10^{-5}	10^{-3}	10^{-3}
Loss function	MSE_{KM}	MAE	MAE

Abbreviations: LR, learning rate (10^{-3} , 10^{-4} , and 10^{-5}); MSE_{KM} , physics-informed mean squared error; N_{HL} , number of hidden layers (3, 5, 7, and 9); N_{NR} , number of neurons (5, 50, 100, and 150).

3.1 | Simulation study

Figure 2 shows the NMAE, NRMSE, and CV_{Net} values for the noise level. The supervised DNN had the highest accuracy and noise immunity for CBF and ATT estimations, owing to the lowest NMAE and NRMSE. Although CBF estimated using the unsupervised DNN had a smaller NMAE and NRMSE than that estimated using the conventional method, this relationship was reversed for ATT. In addition, the supervised DNN showed a higher network precision for CBF estimation than the unsupervised DNN because of the lower CV_{Net} . Conversely, the network precisions of the two DNNs for ATT estimation were not significantly different.

Figure 3 presents the Bland–Altman plots between the ground truth and the predicted CBF. At an SD of 0% M_0 (i.e., no noise), while CBF estimation was slightly inaccurate with both DNNs, the conventional method estimated CBF with complete accuracy. There was no clear difference in CBF estimation between the methods at an SD of 0.02% M_0 . Conversely, with an SD of 0.2% M_0 , the estimation characteristics of the methods differed significantly. The supervised DNN exhibited no bias and negative proportionality. However, the unsupervised DNN and conventional method exhibited similar characteristics, which differed from those of the supervised DNN. These two methods showed a positive bias and slightly positive proportionality. Notably, the conventional method exhibited the largest bias among the analyzed methods.

Figure 4 shows the relationship between CBF estimation accuracy and the ground truth pairs of CBF and ATT. Even with an SD of 0% M_0 , CBF estimation accuracy with the unsupervised DNN was slightly decreased, with a low CBF and long ATT. With an SD of 0.02% M_0 , all methods overestimated CBF when the ground truth CBF was small. However, no relationship was observed between CBF estimation accuracy and ground truth ATT. Moreover, consistent with the results of the Bland–Altman analyses, the supervised DNN had the highest accuracy among the methods, with an SD of 0.2% M_0 , indicating high noise immunity.

Figure 5 shows the Bland–Altman plots between the ground truth and the predicted ATT. Like CBF, at an SD of 0% M_0 , the conventional method estimated ATT with 100% accuracy. Furthermore, while the supervised DNN estimated ATT accurately, the unsupervised DNN presented negative proportionality. As the noise increased, negative proportionality was observed in all methods, but a limited change was observed in the biases.

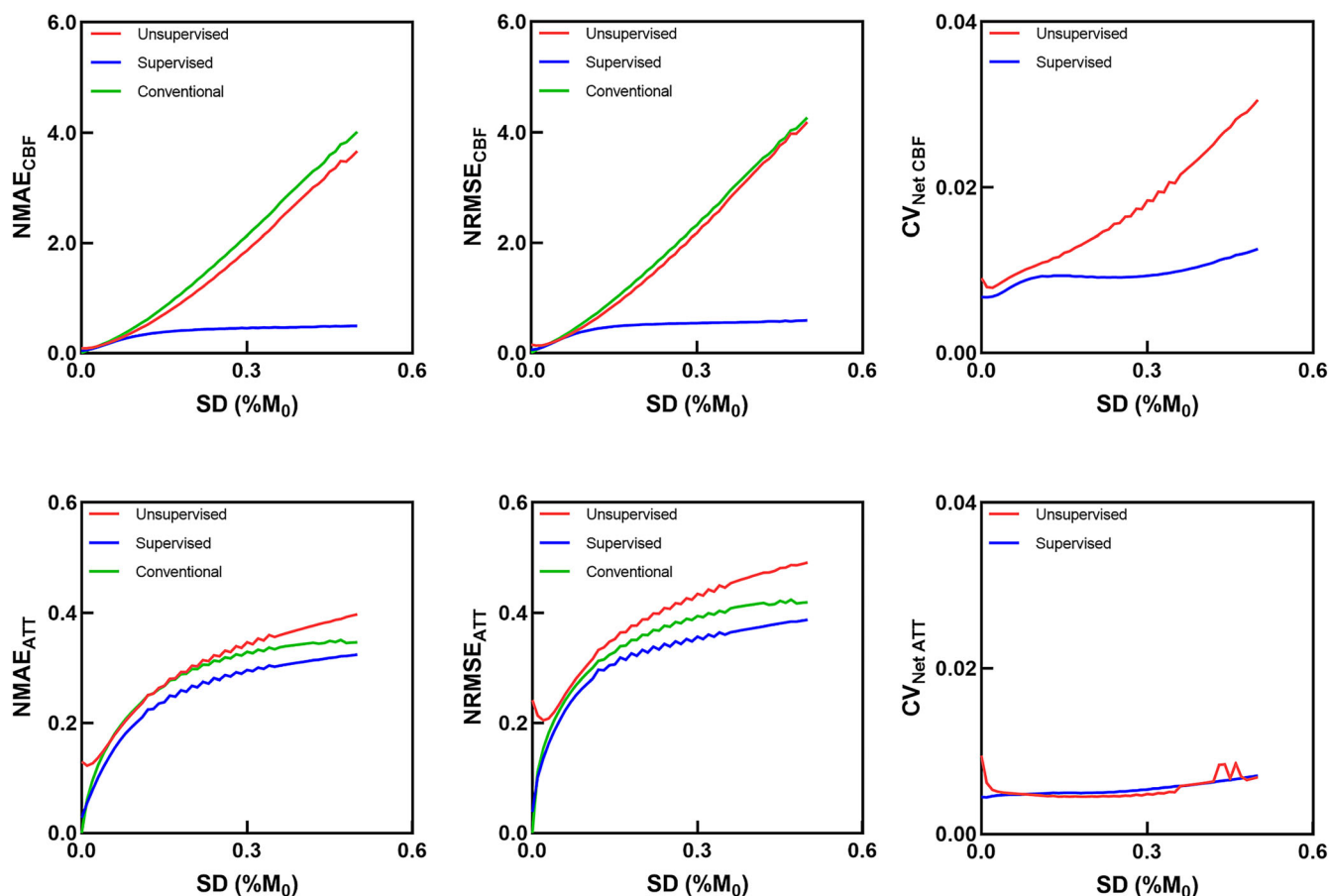


FIGURE 2 Changes in the NMAE (left column), NRMSE (middle column), and coefficient of variation of the network (CV_{Net} ; right column) for the standard deviation of the artificial noise. The CV_{Net} of the conventional method is zero because it is deterministic. The upper and lower rows present CBF and ATT, respectively.

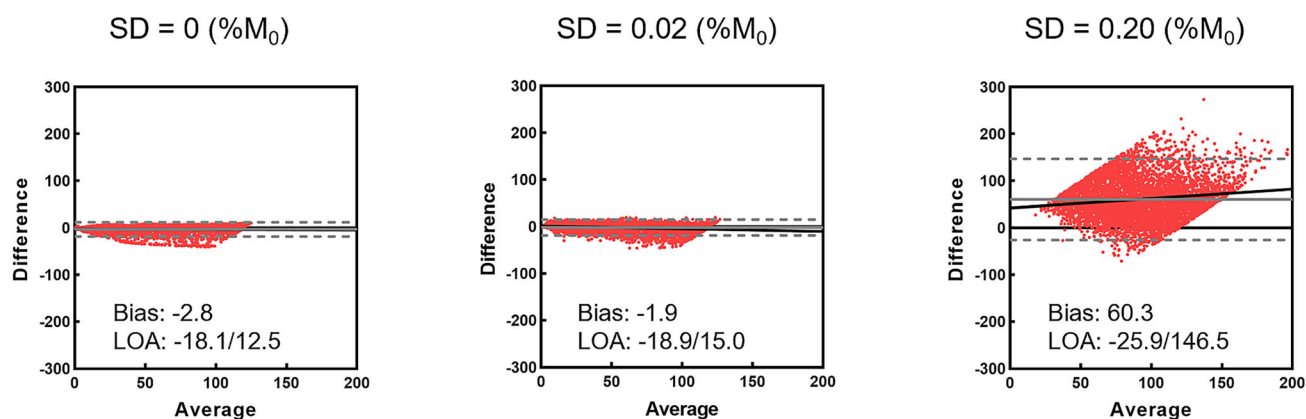
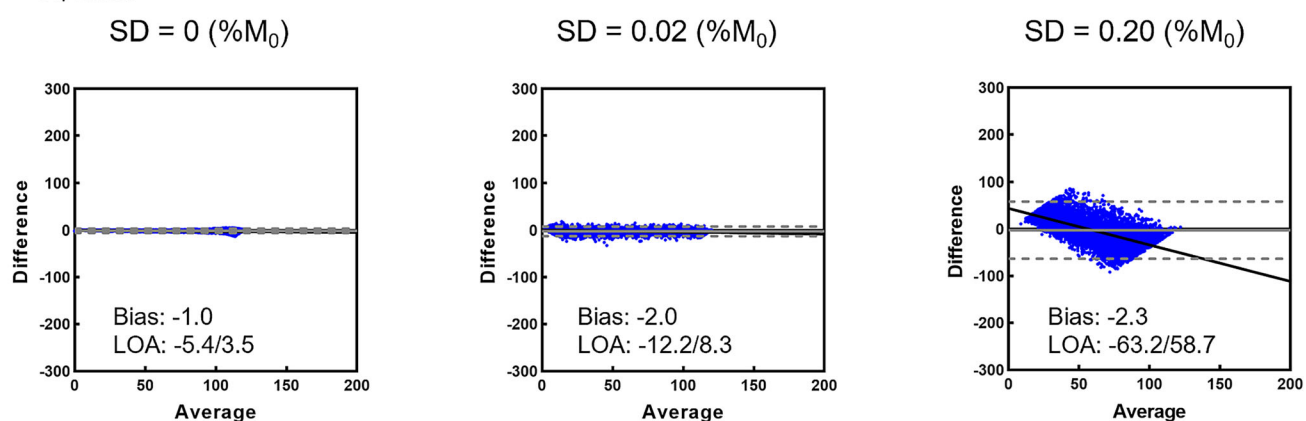
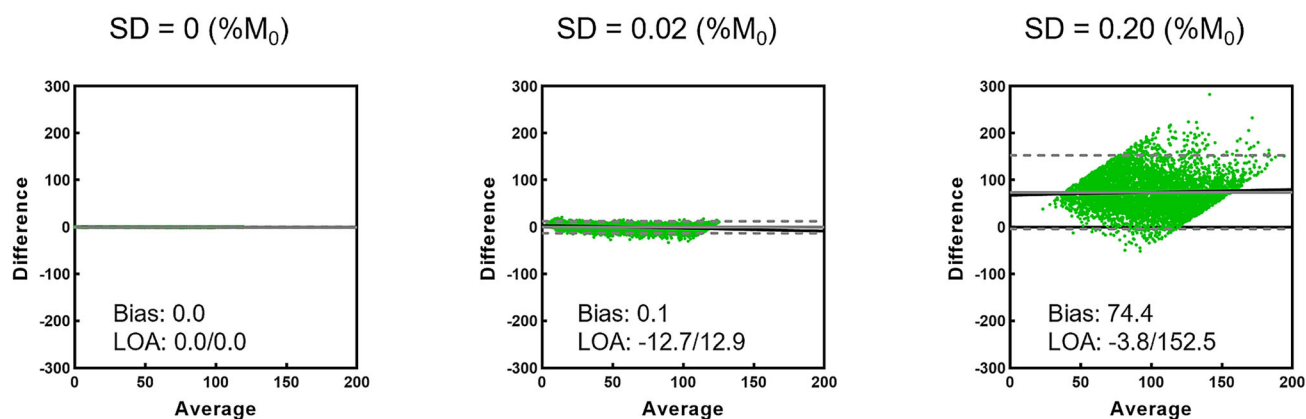
CBF_{Unsupervised}CBF_{Supervised}CBF_{Conventional}

FIGURE 3 Bland–Altman plots between the ground truth and predicted CBF values at SDs of 0% M_0 (left column), 0.02% M_0 (middle column), and 0.20% M_0 (right column). The upper, middle, and lower rows show unsupervised learning, supervised learning, and the conventional method, respectively. Bias and limits of agreement (LOAs) are also presented in the graph area. Black solid lines show regression lines. Gray solid and dashed lines present the bias and LOA (mean $\pm 1.96 \times$ SD), respectively.

Figure 6 shows the relationship between ATT estimation accuracy and ground truth pairs of CBF and ATT. At an SD of 0% M_0 , the supervised DNN and conventional method accurately estimated ATT. However, accuracy was compromised with the unsupervised DNN under low CBF conditions. Even at an SD of 0.02% M_0 , the estimation performances of the supervised DNN and the conventional method were negatively affected: short ATTs were overestimated under low CBF conditions. Consistent with the results of the Bland–Altman analyses, at an SD of 0.2% M_0 , all methods over- and under-estimated short and long ATT, respectively.

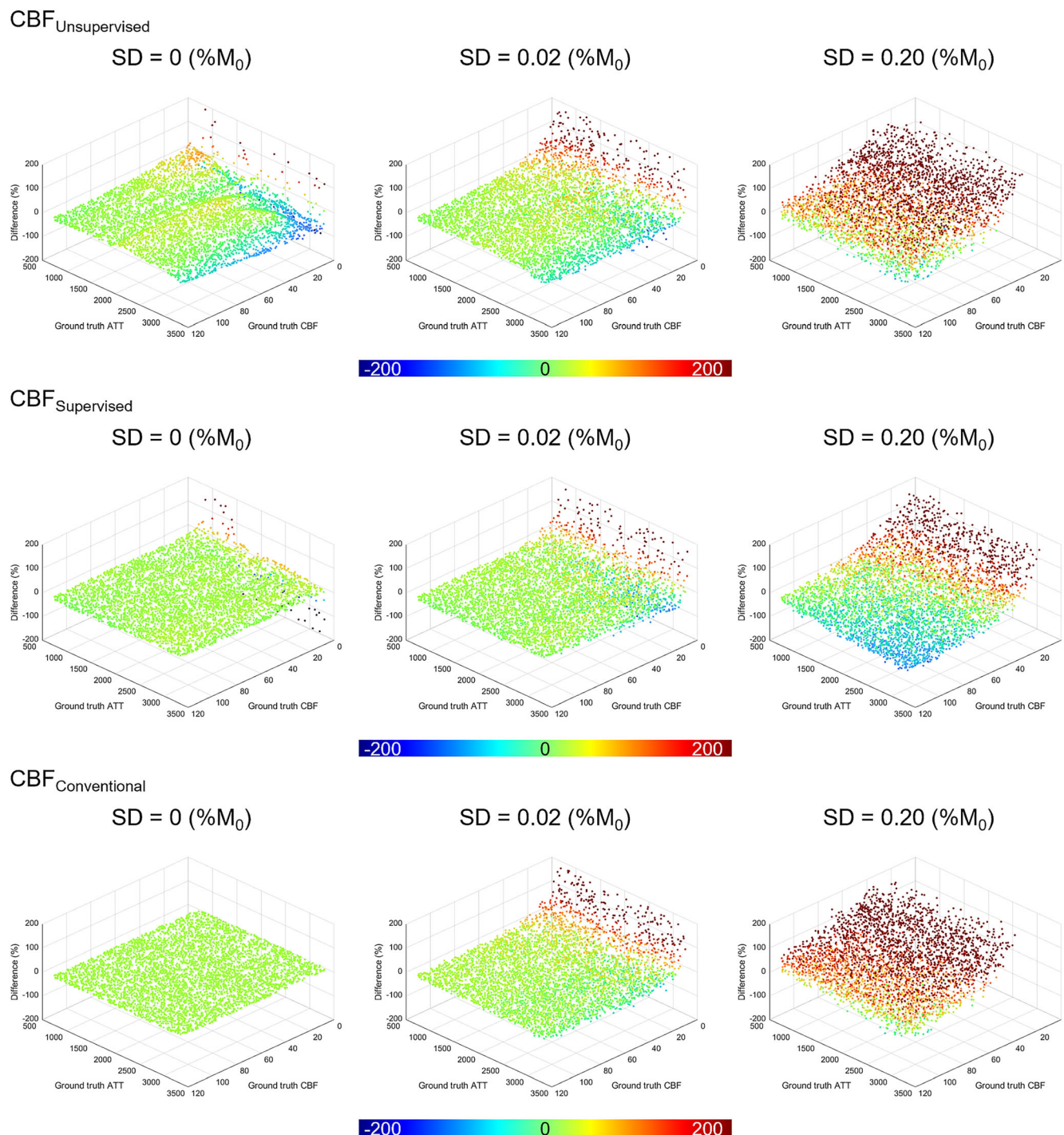


FIGURE 4 Relationship between the prediction accuracy of CBF and ground truth pairs of CBF and ATT using the test data under three noise levels (SD of 0, 0.02, and 0.20 $\%M_0$). Prediction accuracy is calculated as a percent difference, where the difference between the predicted and ground truth CBF values is divided by the ground truth CBF. The upper, middle, and lower rows show unsupervised learning, supervised learning, and the conventional method, respectively.

3.2 | In vivo study

Figure 7 shows the results of CBF measurement in the in vivo data using the simulation-based DNN and the conventional method. CBF estimated using the supervised DNN changed slightly even when the noise increased (Figure 7A). This was consistent with the measurement results (Figure 7C,D). Moreover, CBF estimated at 0 $\%M_0$ using the supervised DNN was the lowest among those with all other methods (Figure 7B).

Figure 8 shows the results of ATT measurement in the in vivo data using the simulation-based DNN and the conventional method. In contrast to CBF, the increase in relative ATT with noise was the smallest with the unsupervised DNN and the largest with the conventional method

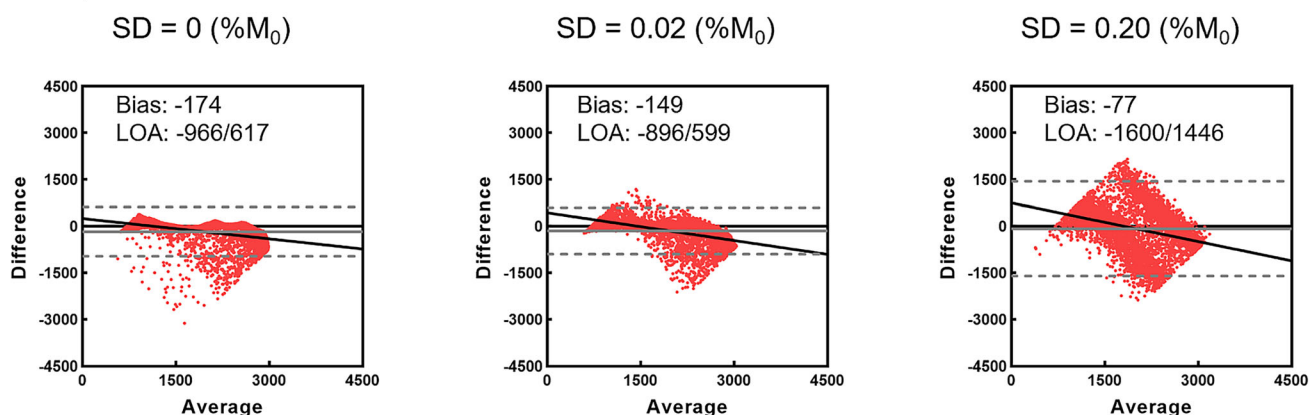
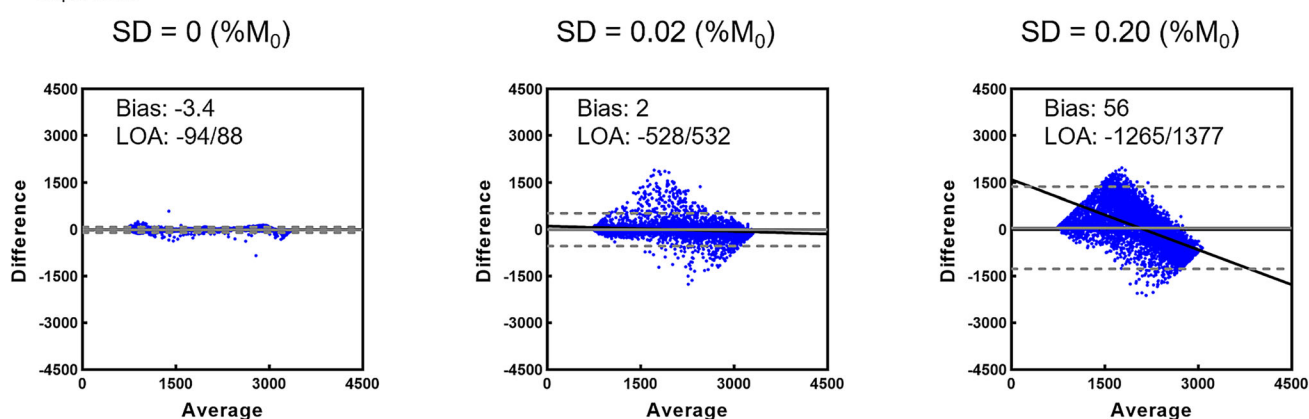
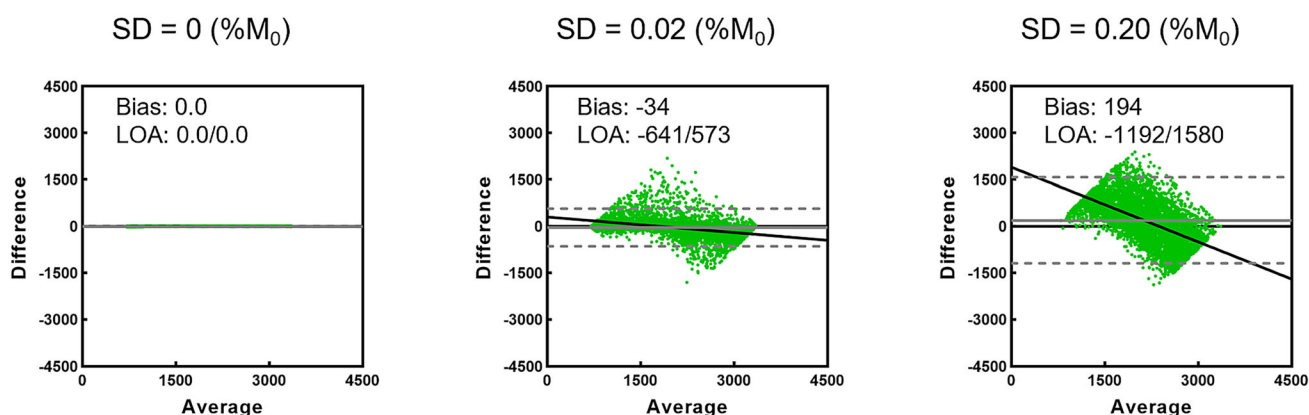
ATT_{Unsupervised}ATT_{Supervised}ATT_{Conventional}

FIGURE 5 Bland-Altman plots between the ground truth and predicted ATT values at SDs of 0% M_0 (left column), 0.02% M_0 (middle column), and 0.20% M_0 (right column). The upper, middle, and lower rows show unsupervised learning, supervised learning, and the conventional method, respectively. Bias and LOAs are also presented in the graph area. Black solid lines show regression lines. Gray solid and dashed lines present the bias and LOA (mean $\pm$ 1.96 SD), respectively.

(Figure 8A,C). Similar to CBF, Figure 8B shows that ATT estimated at 0% M_0 using the supervised DNN was the smallest among those with all other methods.

Figures 9 and S2 show the results of comparison among the unsupervised DNN trained using in vivo data, unsupervised DNN trained using simulation data, supervised DNN, and conventional method. For the unsupervised DNN trained using in vivo data, while using in vivo data for DNN training did not improve the performance of CBF estimation, it improved the noise immunity of ATT estimation compared to that trained using simulation data.

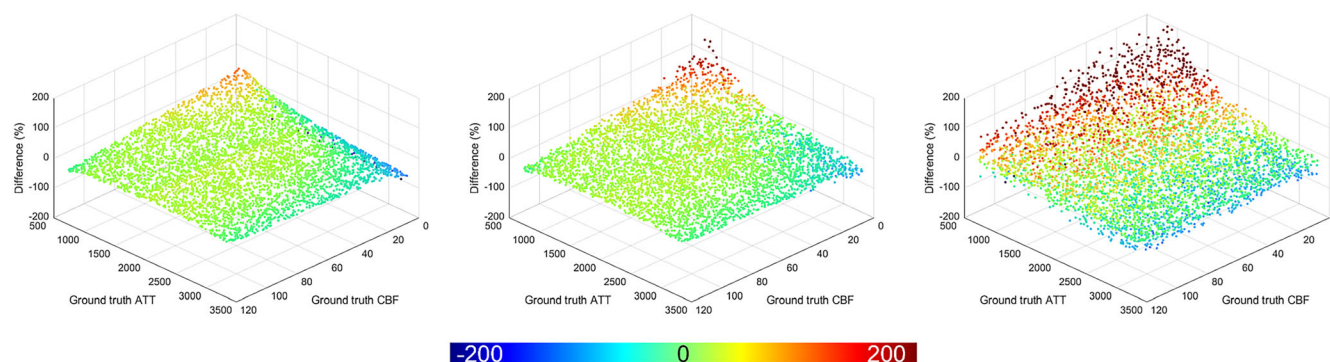
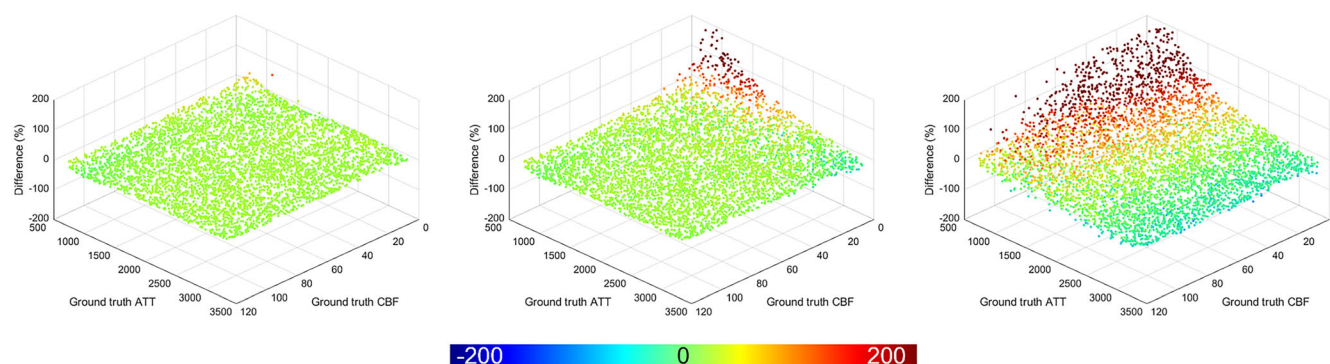
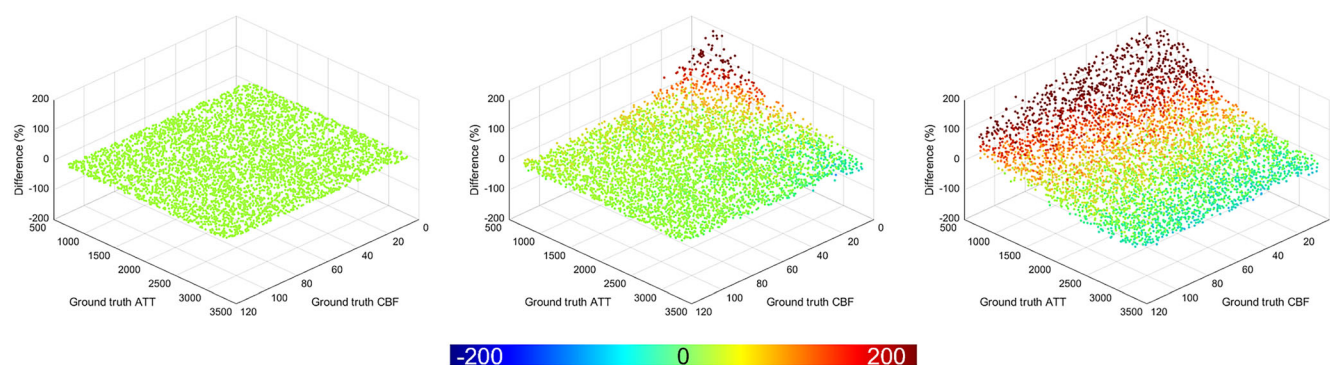
ATT_{Unsupervised}SD = 0 (%M₀)SD = 0.02 (%M₀)SD = 0.20 (%M₀)ATT_{Supervised}SD = 0 (%M₀)SD = 0.02 (%M₀)SD = 0.20 (%M₀)ATT_{Conventional}SD = 0 (%M₀)SD = 0.02 (%M₀)SD = 0.20 (%M₀)

FIGURE 6 Relationship between the prediction accuracy of ATT and ground truth pairs of CBF and ATT using the test data under three noise levels (SD of 0, 0.02, and 0.20% M₀). Prediction accuracy is calculated as a percent difference, where the difference between the predicted and ground truth ATT is divided by the ground truth ATT. The upper, middle, and lower rows show unsupervised learning, supervised learning, and the conventional method, respectively.

4 | DISCUSSION

Physics-informed unsupervised learning was implemented to estimate CBF and ATT from multi-delay ASL signals, and its performance was compared with supervised learning and the conventional method using simulation and in vivo data. The simulation study showed that the supervised DNN had the highest accuracy and noise immunity for CBF and ATT estimations. The in vivo study showed that the physics-informed unsupervised DNN had higher noise immunity in CBF and ATT estimations than did the conventional method. Conversely, the unsupervised DNN exhibited lower and higher noise immunity in CBF and ATT estimations, respectively, than the supervised DNN. Additionally, the use of

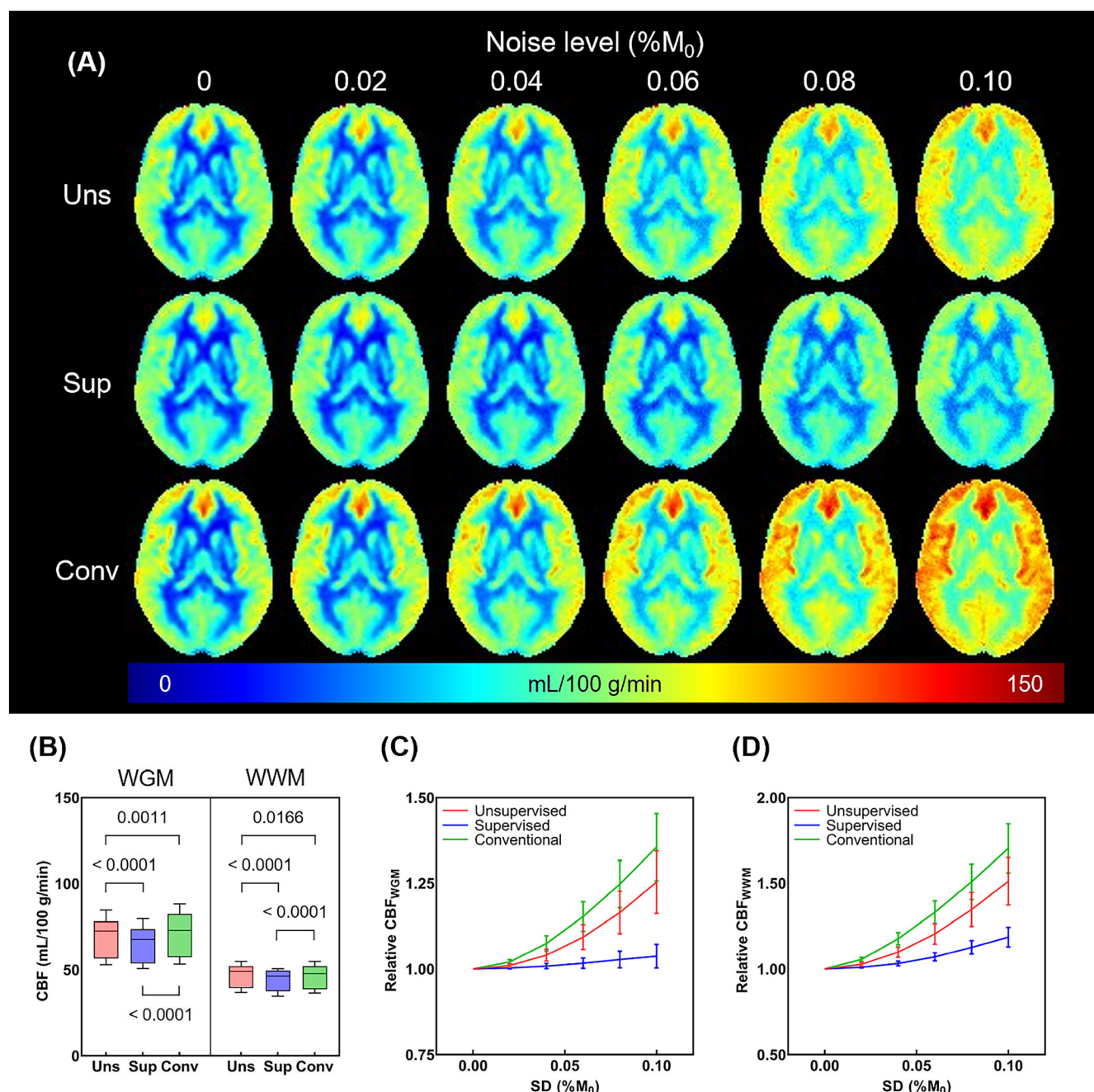


FIGURE 7 A, Averaged CBF maps for all subjects under different noise levels. The upper, middle, and lower rows show unsupervised learning (Uns), supervised learning (Sup), and the conventional method (Conv), respectively. B, CBF values using each method measured in the WGM (left) and WWM (right). C, D, Changes in relative CBF values for the SD of the artificial noise in WGM (C) and WWM (D).

in vivo data for training improved the performance of the unsupervised DNN. To the best of our knowledge, this is the first report to demonstrate the feasibility of physics-informed unsupervised learning for CBF and ATT estimations using multi-delay ASL and compare its performance with that of supervised learning. The findings of this study would contribute to developing accurate and precise parameter-estimating DNNs, facilitating the clinical use of multi-delay ASL.

4.1 | Network architecture

This study adopted a parallel network²³ consisting of two subnetworks with the same N_{HL} and N_{NR} . Ideally, the subnetworks should be optimized individually, because the optimal N_{HL} and N_{NR} may differ between CBF and ATT estimations. However, it is difficult to evaluate the numerous

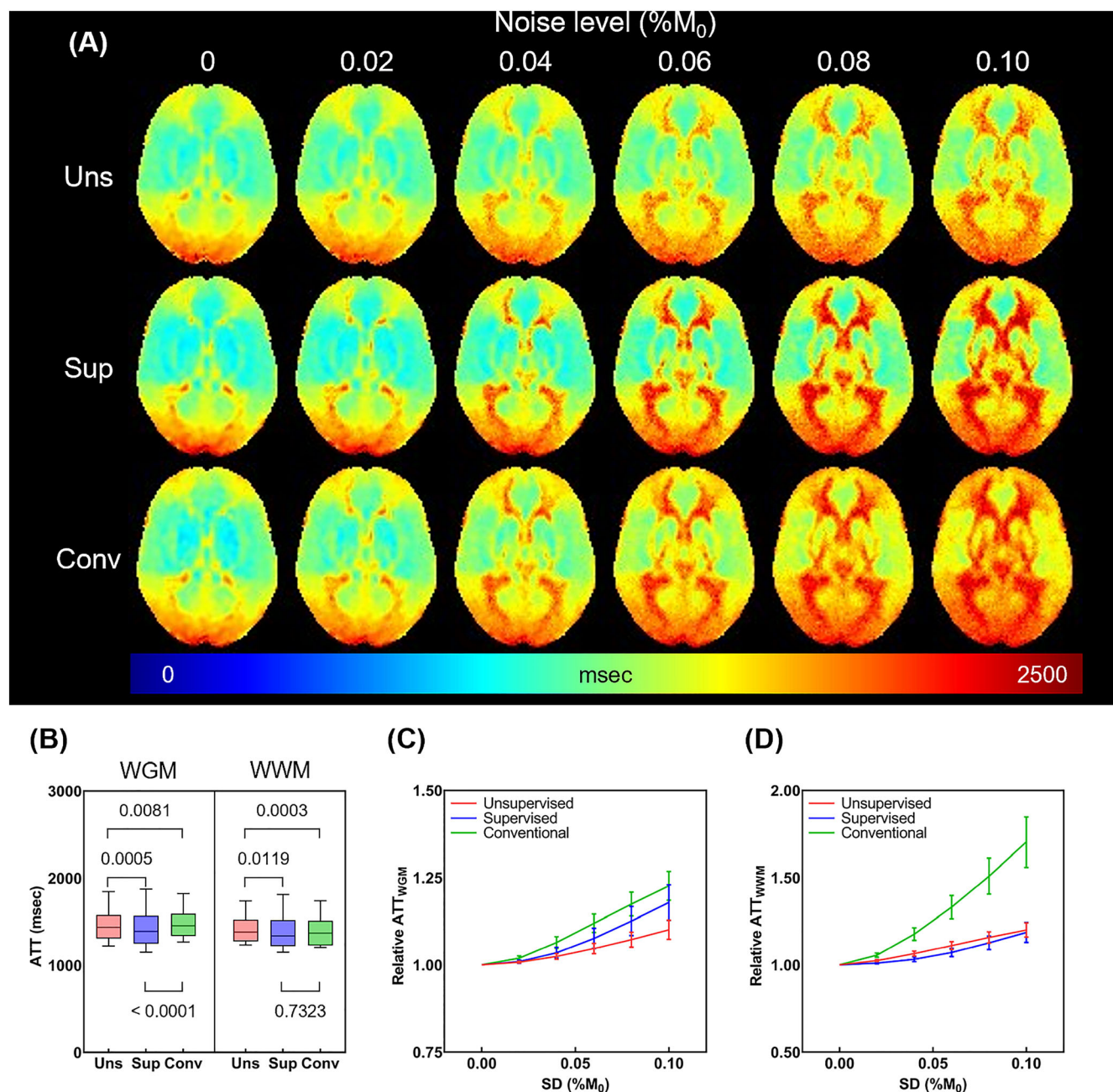


FIGURE 8 A, Averaged ATT maps for all subjects under different noise levels. The upper, middle, and lower rows show unsupervised learning (Uns), supervised learning (Sup), and the conventional method (Conv), respectively. B, ATT values using each method measured in the WGM (left) and WWM (right). C, D, Changes in relative ATT values for the SD of the artificial noise in WGM (C) and WWM (D).

combinations of hyperparameters. Because currently, we cannot optimize DNN architectures a priori, this study attempted to test as many combinations as possible during the preliminary grid search. The optimized architectures analyzed in this study were not ideal because of the limited number of search combinations. However, satisfactory performances of both DNNs indicated the absence of any major problem in the optimization process.

In this study, the network precision (CV_{Net}) was evaluated because DL approaches are stochastic processes. In particular, the network precision for CBF estimation was higher with supervised learning than with unsupervised learning, whereas that for ATT estimation was not significantly different. These results recommend the use of supervised learning for parameter estimation using multi-delay ASL because reproducible estimation is a desirable property of DL.

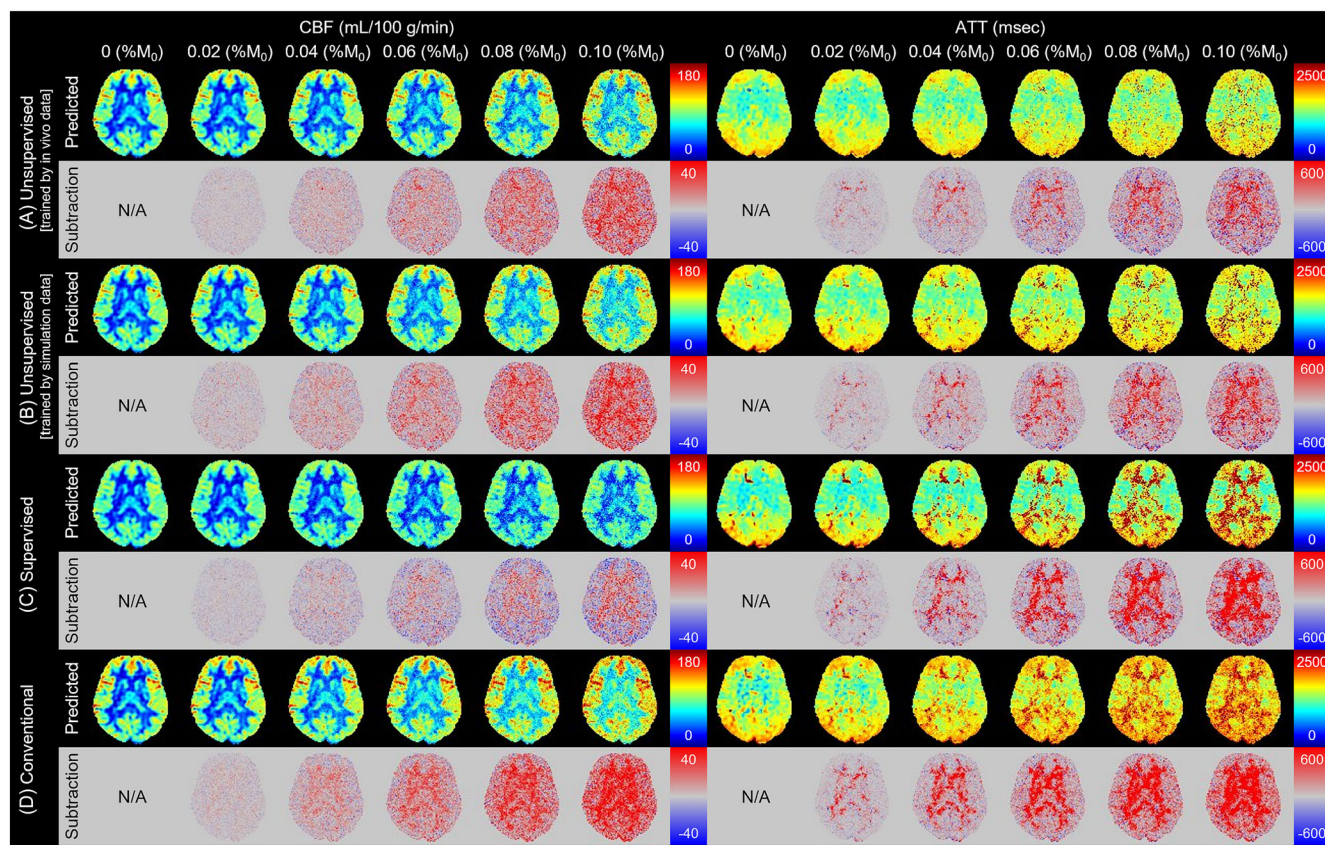


FIGURE 9 A representative case (22-year-old male) of CBF (left panel) and ATT (right panel) maps predicted by the unsupervised network trained using in vivo data (A), the unsupervised network trained using simulation data (B), the supervised network (C), and the conventional method (D). Odd and even rows show predicted maps and subtraction results between the original (i.e., 0% M_0) and each level of synthesized noisy images, respectively.

4.2 | Simulation study

The inaccuracy of some CBF estimations with the unsupervised DNN at 0% M_0 might be caused by ATT estimation inaccuracy because the performance of the unsupervised DNN is affected by both CBF and ATT estimations. This originates from the PILF mechanism, which calculates ASL signals from both predicted CBF and ATT. Furthermore, similar CBF estimation characteristics were observed between the unsupervised DNN and the conventional method, which may arise from their estimation processes. The PILF minimizes the difference between the observed and predicted signals using MSE, whereas the conventional method estimates the CBF using LS. Therefore, although the use and nonuse of a DL framework introduces considerable differences, the PILF and conventional method have an essentially similar process in parameter estimation. Conversely, as the noise increased, the unsupervised DNN and conventional method showed a large positive bias, whereas the supervised DNN showed no bias. This finding suggests that the supervised DNN is effective in mitigating Rician noise biases, as higher noise levels lead to elevated CBF values.³⁰ Therefore, all evaluations associated with the performance of CBF estimation showed that the supervised DNN had the highest accuracy and noise immunity in CBF estimation. In particular, changes in NMAE and NRMSE with the noise level were significantly lower with the supervised DNN than with others. Moreover, the negative proportionality of the supervised DNN does not indicate a compromised accuracy, but rather the preserved accuracy of the supervised DNN, as some CBF ranges are estimated accurately even with increased noise.

ATT estimation with the unsupervised DNN presented negative proportionality, despite an SD of 0% M_0 . Similar to CBF, this may stem from the characteristics of PILF as shown in Figure 4, which indicates the compromised accuracy of ATT estimation with low CBF. Inaccurate ATT estimations owing to low CBF led to high NMAE and NRMSE under low SD conditions, as shown in Figure 2. However, the compromised ATT accuracies may be attributable to the characteristics of the ATT itself rather than the estimation techniques, because the results of the Bland-Altman analyses showed that all methods behaved similarly for noise increase. While CBF is a scaling factor of the kinetic curve, ATT is the time at which the signal rises from the baseline (i.e., zero) and defines the shape of the kinetic curve. Rician noise obscures the position of the baseline, which distorts the ATT estimation. However, none of these methods could fundamentally solve the problem of ATT estimation under noisy conditions. Nonetheless, as shown in Figure 2, the supervised DNN had the highest accuracy and noise immunity in ATT estimation.

For both CBF and ATT, although the DNNs outperformed the conventional method under noisy conditions, this relationship was reversed under no-noise conditions, where CBF and ATT were estimated with 100% accuracy with the conventional method. A previous study has reported similar trends.²³ These results may be due to the stochastic process of DL training, but further studies are required to clarify this. However, the DNNs outperform the conventional method in practical use because ASL has an inherently poor SNR, as mentioned previously.

4.3 | In vivo study

Noisy images were synthesized by adding Rician noise to the original images. The brain regions in the original images had sufficient ASL signals, in which the Rician distribution could approximate a Gaussian distribution.³¹ In addition, Gaussian distribution has reproductive properties. Therefore, there seems to be no major problem in the synthesis of noisy images. Furthermore, in this study, the noise levels of the training and in vivo data were set to 0%–0.5% M_0 and 0%–0.1% M_0 , respectively. As ASL signals are less than 1.0% of the equilibrium magnetization of brain tissue (i.e., 1.0% M_0),³² the worst SNR of the training data was less than 2. In a previous study, the SNR of the images used in this study was approximately 20 for WGM.²¹ Although SNR of ASL signals cannot be defined strictly because the imaging conditions (for example, LD and PLD) and physiological parameters (for example, CBF, ATT, T_{1a} , and $T_{1\rho}$) comprehensively affect the ASL signals, the current noise level setting for training data generation sufficiently included the noise levels of the original and synthesized noisy images. The use of simulation data is considered a practical and reasonable strategy for training unsupervised DNNs^{22,23} because numerous images with a wide range of noise levels are required for direct learning from measured data.

Among the assessed methods, CBFs estimated at 0% M_0 using the supervised DNN were the smallest in the WGM and WWM. These results indicate that the supervised DNN is the most accurate in CBF estimation because the simulation results showed that the unsupervised DNN and the conventional method had large positive biases under noisy conditions. This also indicates that the supervised DNN had the highest noise immunity to the noise originally present in the original images. In addition, consistent with the simulation results, the unsupervised DNN showed smaller CBF than the conventional method, which indicates a higher accuracy of the unsupervised DNN than the conventional method. The relative change in CBF also showed that all CBF values increased with an increase in noise.^{21,30} However, the CBF estimated using the supervised DNN changed slightly, and its rate of increase was the smallest, indicating that the supervised DNN had the highest noise immunity among all methods.³⁰ Therefore, in terms of accuracy and noise immunity, the supervised DNN exhibited the best performance in CBF estimation.

Despite slight differences, the ATT estimated at 0% M_0 using the supervised DNN was significantly smaller than that estimated using other methods, indicating the highest accuracy of the supervised DNN because all ATT values increased with increasing noise. Conversely, the increase in relative ATT of the WGM was smallest with the unsupervised DNN, indicating the highest noise immunity of the unsupervised DNN. Unlike the results of CBF measurement, a discrepancy between the results of accuracy and noise immunity was observed for ATT measurement. The exact reason for this is unclear, but the compatibility between the added noise and unsupervised learning may be the reason. While the synthesized images included original and artificial noise, the simulation data used for training only included artificial noise. In particular, although the same noise distribution was assumed between the actual images and the simulation data, the unsupervised DNN learned information on artificial noise directly from the simulation data, which led to a higher adaptability of unsupervised learning to the added noise and might have led to higher noise immunity. Further studies are required to clarify the exact reason for these findings. However, the comparison results between the unsupervised DNN trained using in vivo and simulation data might support this inference, because the unsupervised DNN trained using in vivo data exhibited improved noise immunity compared with that trained using simulation data.

In contrast, if the above discussion is true, the results of the in vivo ATT analysis can be considered promising because direct learning from noisy data using the physics-informed unsupervised DNN can provide higher noise immunity and specificity to the noise characteristics in the actual images. Moreover, although this study used only temporal noise information, image-based regression³³ such as convolutional neural networks³⁴ or U-nets³⁵ can use spatial noise information. A combination of PILF and image-based regression will enable an improvement in the accuracy of ASL parameter estimation, because image-based regression cannot use the ground truth. Furthermore, based on the results of this study, a hybrid of supervised and unsupervised learning for CBF and ATT estimation may provide a higher performance in ASL parameter estimation.

The design of the study should be discussed. Although an advantage of physics-informed unsupervised learning is the direct learning of signal characteristics from measured data, the results of this study show that the overall performance of supervised learning was higher than that of physics-informed unsupervised learning. The main reason for this may be the consistent use of simulation data for the training and optimization of DNNs and evaluation of their performance. In this scenario, the current study design may be advantageous for supervised learning because, while supervised learning can always learn from correct values and minimize the difference between the predicted and ground truth values directly, unsupervised learning predicts CBF and ATT indirectly by minimizing the difference between the observed and predicted ASL signals. However, even under these conditions, physics-informed unsupervised learning showed promising results, which sufficiently demonstrates the feasibility of this technique. Obviously, a physics-informed unsupervised DNN trained using measured signals is desirable; however, this is a topic for future studies because we do not have sufficient data to train unsupervised DNNs. Therefore, we used original and synthesized noisy images to train an unsupervised DNN. We also attempted to train a DNN using only the original image data, but this attempt was not successful, which

may be due to the insufficient amount of data available to train the DNN. However, similar to this study, in previous studies that used simulation data to train unsupervised networks for intravoxel incoherent motion parameter estimation, they outperformed the conventional techniques.^{22,23} Therefore, although the performance of physics-informed unsupervised learning could not be sufficiently enhanced owing to the lack of measured signals for training, the results of this study are considered promising.

This study had several limitations. First, this was a single-center, single-scanner trial with a small cohort that included only healthy adults who were relatively young. Second, the clinical utility of the study results should have been evaluated for various diseases (such as cerebrovascular diseases, neurodegenerative diseases, brain tumors, and psychiatric disorders). Third, additional investigations are required to verify the generalizability of the PILF to other ASL acquisition designs^{13,15,16} and KMs^{32,36–38} because only one ASL acquisition design and a single-compartment model were used in this study. Fourth, the hyperparameters of the unsupervised network trained using in vivo data should be optimized exclusively for it, as they were identical to those of the unsupervised network trained using simulation data. Finally, the CBF values should be validated by comparison with those obtained using ¹⁵O-water PET.³⁹

5 | CONCLUSIONS

The physics-informed unsupervised learning implemented in this study could estimate CBF and ATT from multi-delay ASL signals. The physics-informed unsupervised DNN showed higher accuracy and noise immunity in ASL parameter estimation than the conventional method, which used the WD method and LS solution for estimating ATT and CBF, respectively. The supervised DNN demonstrated a higher accuracy in CBF and ATT estimation and higher noise immunity in CBF estimation than the physics-informed unsupervised DNN. Conversely, the physics-informed unsupervised DNN exhibited superior noise immunity in ATT estimation. The accuracy of supervised learning is only affected by the parameters to be estimated. By contrast, the accuracy of physics-informed unsupervised learning is affected by both CBF and ATT.

ACKNOWLEDGMENTS

This study was supported in part by JSPS KAKENHI (grant numbers 21K15802 and 21K07616).

CONFLICT OF INTEREST STATEMENT

Naoyuki Takei is an employee of GE HealthCare.

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How to cite this article: Ishida S, Fujiwara Y, Takei N, Kimura H, Tsujikawa T. Comparison between supervised and physics-informed unsupervised deep neural networks for estimating cerebral perfusion using multi-delay arterial spin labeling MRI. *NMR in Biomedicine*. 2024;37(10):e5177. doi:[10.1002/nbm.5177](https://doi.org/10.1002/nbm.5177)